

Introduction

This guide explains the core features of our custom Odoo module, *TRMNL*. The source code is available at: https://github.com/Abilium-GmbH/pse_trmnl_odoo_connector

TRMNL enables Odoo to be used as an API server for TRMNL e-ink displays. It allows users to connect their devices to Odoo and display Odoo data on them.

This module was developed as part of the Software Engineering Lab at the University of Bern, in cooperation with Abilium GmbH. It is not affiliated with TRMNL Holdings LLC.

Requirements

The module is designed for Odoo 19 and is compatible with TRMNL firmware v1.8.2, the latest at the time of writing. A running Odoo instance is required.

External Python dependencies:

- pillow (version 12.1.1)
- requests (version 2.32.5)

Note: The repository README.md contains useful information if you plan to run Odoo in Docker (PostgreSQL, first-time setup, and daily operation).

For **local or Docker installations**, TRMNL devices must be able to reach your Odoo server over the network. Set the system parameter `web.base.url` to an address the device can access (for example <http://192.168.x.x:8069>), not localhost or 127.0.0.1. On the first successful device poll, the module may also auto-update `trmnl.public_base_url` from the device's request so image URLs stay reachable on your LAN.

Device Pairing

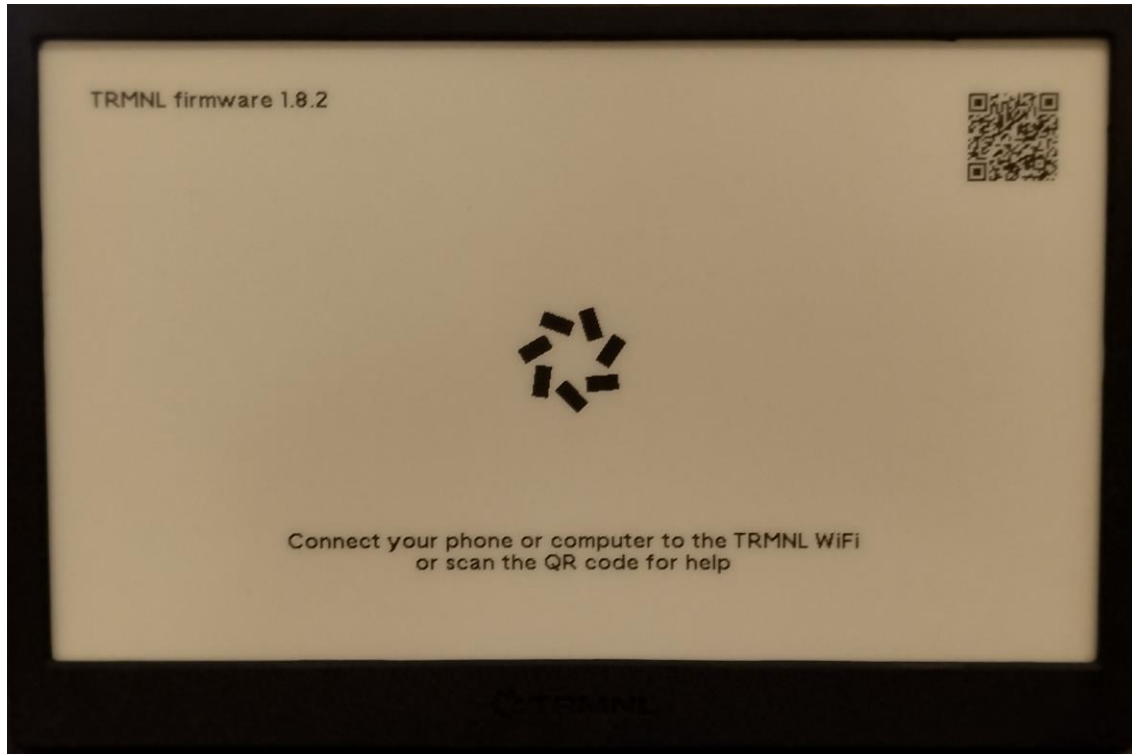
Module Installation

1. In Odoo, go to Apps.

2. Remove the "Apps" filter from the search bar (the module is classified as a module, not a standalone app).
3. Search for "TRMNL" and install the module.

Device configuration

1. Turn on the device by flipping the switch on the back.
2. Press and hold the button on the back for 5–10 seconds. When you release it, the screen should refresh and display instructions to connect to the TRMNL Wi-Fi.



3. Connect to the TRMNL network from your device's available Wi-Fi networks. A device configuration page should open in your web browser.

Wi-Fi Configuration

RVN-79102 2.4G	📶 🔒
JOOWIN-2G&5G 2.4G	▼ 🔒
DIRECT-B9-HP Smart Tank 7300 2.4G	▼ 🔒
RajNet 2.4G	▼ 🔒
Kesar Rest 2.4G	▼ 🔒
Comedor.v, 2.4G	▼
Sunrise_1102471 2.4G	▼ 🔒
dormitorio 2.ynm, 2.4G	▼
Du kommst hier nicht rein 2.4G	▼ 🔒

SSID

Password

Connect
Refresh

Advanced

MAC: 80:F1:B2:80:52:48

4. To connect to our module, first delete any existing device credentials:
 - Scroll down and click Advanced.
 - Click Soft Reset to remove stored credentials (this is required to connect to our module).
 - Confirm the reset in the pop-up by clicking Yes.



Advanced Configuration

Soft Reset
Custom Server

Run Sensor Test

Back to Wi-Fi

5. The device will disconnect during the reset. Reconnect to the TRMNL network and return to Advanced Configuration.
6. Click Custom Server and confirm in the pop-up by clicking Yes.



Advanced Configuration

API Server

Enter your custom server without a trailing / (e.g. <https://trmnl.app>)

Soft Reset

Custom Server

Run Sensor Test

Back to Wi-Fi

7. Enter your server address.

- **Odoo.sh:** use your instance's public HTTPS URL, e.g. <https://<your-instance-name>.odoo.com>
- **Local / Docker:** use the LAN IP and port of your Odoo host, e.g. <http://192.168.1.50:8069>

Then click **Back to Wi-Fi**.

8. The API Server field should now be visible in the Wi-Fi Configuration menu.

RajNet 2.4G	⌵
Kesar Rest 2.4G	⌵
Comedor.v 2.4G	⌵
dormitorio 2.ynm 2.4G	⌵
Du kommst hier nicht rein 2.4G	⌵
Sunrise_1102471 2.4G	⌵
FRITZ!Box 5530 MW 2.4G	⌵
Guest Aligro 2.4G	⌵
HTC Desire 12+ 2.4G	⌵

SSID

Enter your Wi-Fi name

Password

Enter your Wi-Fi password

API Server

<https://<your-instance-name>.odoo.com>

Connect Refresh

Advanced

MAC: 80:F1:B2:80:52:48

Select your Wi-Fi network from the drop-down, enter the network password, and click Connect.

The device connection can be unstable; you may need to reconnect to the TRMNL network manually if disconnected.

9. The device may display a temporary instruction to visit trmnl.com/start. This message is hard-coded in the firmware and can be ignored; wait for the screen to refresh automatically or press the back button to refresh manually.



10. If you see the Abilium logo after the first screen refresh, setup was successful and the device is ready.



Device management

Access

Open TRMNL in the main menu.

Menu	Purpose
TRMNL → Devices	Registered displays, telemetry, refresh rate, logs
TRMNL → Profiles	Configure what Odoo data is rendered for each device
TRMNL → Display Policy	Global rule for unknown or mismatched devices

How devices appear in Odoo

When a TRMNL is configured with your Odoo server and connects over Wi-Fi:

Endpoint	Purpose
GET /api/setup	First registration. Odoo creates a device record and marks it Accepted immediately. The device receives an API key in the setup response.
GET /api/display	Periodic polls for the current image URL, filename, and refresh rate.
POST /api/log	Telemetry and firmware logs, stored on the device's Logs tab.

You **cannot** create devices manually from the list (**New** is disabled). Records appear when hardware contacts the server.

Depending on the **Display Policy** (see below), a device that polls /api/display before a successful setup may appear as **Unknown Device** or **Token Mismatch** until you accept it.

Device List view

TRMNL → Devices shows:

Column	Description
(drag handle)	Manual sort order (sequence). Drag rows to reorder.
Device Name	Friendly label (editable on the form)
MAC Address	Hardware ID (read-only). Canonical format: AA:BB:CC:DD:EE:FF.
Status	Badge: Accepted (green), Token Mismatch (yellow), Unknown Device (red).
Image Filename	Current image filename sent to the device (read-only). Changes when a new PNG is served.
Last Seen At	Last contact with any TRMNL API endpoint.
Refresh Rate (min)	How often the server tells the device to poll (editable, 1–30 minutes). Default: 1 minute.
Battery %	Last reported battery level.
Signal Strength	Wi-Fi quality bucket derived from RSSI (Excellent / Good / Fair / Poor).

Search filters: **Accepted**, **Needs Attention** (token mismatch or unknown device), group by **Status**.

List actions (select rows → **Action** menu):

Action	Effect
Accept Selected	Manually accept devices that have presented a token (only meaningful when Display Policy is Respond with an error). Skips devices without a stored token.
Remove Selected	Delete device records immediately from Odoo.
Reset Selected	Schedule a factory reset on the device's next poll, then delete the record.

TRMNL Devices Profiles Display Policy							
Devices				Q Search...		1-1/1	
☐	Device Name	MAC Address	Status	Image Filename	Last Seen At	Refresh Rate (min)	Battery Percentage Signal Strength
☐	Office Display	94:A9:90:8B:2A:A0	Accepted	profile_28_20260531T2233...	Jun 1, 12:43 AM	1	12 Good

Device Form View

Open a device from the list to see the full form.

Header buttons:

Button	When to use
Accept Device	Visible when Display Policy is Respond with an error and the device is not yet Accepted . Adopts the token the device last presented.
Reset Device	Sends a factory-reset signal on the next poll (clears Wi-Fi and API key on the device), then removes the Odoo record.
Remove Device	Removes the Odoo record only, or Reset & Remove to reset the physical device first (recommended).

Form sections:

- **Configuration** (top): image URL, refresh rate (minutes), filename.
- **Telemetry**: battery, signal, last reported refresh rate, registration and last-seen timestamps.
- **Logs** (notebook tab): entries from POST /api/log (select and delete individual lines if needed).

TRMNL
Devices
Profiles
Display Policy

Devices
Display

Reset Device
Remove Device
Accepted
Token Mismatch
Unknown Device

Device Name
Display

MAC Address
94:A9:90:8B:2A:A0

CONFIGURATION

Image URL
http://192.168.1.127:8069/api/profile/image?token=goXybkHQUvPjN7UZZjyXoPq68EJL3_APCF8q1vyXmc

Refresh Rate
1
minutes

Image Filename
profile_1_20260601T112654_8cf3ffb16146

TELEMETRY

Battery Percentage
0

Signal Strength
Good

Last Reported Refresh Rate (min)
1

Added At
Jun 1, 1:06 PM

Last Seen At
Jun 1, 1:34 PM

Last API Call
Display

Logs

Device Log ID

Name

Created At

Wi-Fi Status

Refresh Rate

Battery Voltage

Display Policy

TRMNL → Display Policy configures how Odoo responds to **/api/display** requests from:

- devices not yet registered via **/api/setup**, or
- devices whose **Access-Token** does not match the stored token.

It does **not** change normal operation after a successful **/api/setup** (those devices are **Accepted** with a server-issued token).

The setting is stored in the system parameter **trmnl.display_unknown_device_policy**.

Policy	Behavior
Respond with an error (default, recommended)	Device receives an “Unauthorized” placeholder image. Odoo may create or update an Unknown Device or Token Mismatch record. Accept manually when ready.
Automatically accept the device	Odoo adopts the token from that poll, marks the device Accepted , and serves the normal display payload.
Send a reset command	Odoo returns {"status": 500} to the device (firmware factory-reset path) and deletes the device record. Reconfigure Wi-Fi and Custom Server afterward.

This is different from the admin **Reset Device** button on the device form, which sets `reset_firmware: true` on the next poll and then removes the record.

Use **Accept Device** on the device form (or **Accept Selected** in the list) when the policy is **Respond with an error** and a token has been stored from a prior poll.

Profile Management/Renderer

Overview

A Profile defines what Odoo data is drawn into a PNG for a TRMNL device:

1. Choose an Odoo model (e.g. Calendar Events, Tasks, Sales Orders)
2. Pick a View Type (layout)
3. Configure filters, fields, and layout-specific options
4. Odoo renders a 1-bit e-ink PNG (default canvas 800×480; device-reported size is used when available)
5. On each device poll, Odoo serves that image via a secured URL

Each profile is linked to one device. A device can have multiple profiles, but only the lowest-sequence active profile is used for display.

The screenshot shows the TRMNL web interface with a purple header bar containing the logo and navigation links: TRMNL, Devices, Profiles, and Display Policy. Below the header, a 'New' button is visible next to a 'Profiles' tab. The main content area displays a form for creating a new profile, organized into four sections:
1. **IDENTITY**: Includes fields for 'Name', 'Device', and an 'Active' toggle switch which is currently turned on.
2. **DATA SOURCE**: Includes a field for 'Odoo Model' with a help icon.
3. **DEVICE DELIVERY STATUS**: Includes fields for 'Device Last Polled' (with a help icon), 'Next Expected Poll', and 'Device Refresh Rate' (with a help icon and a value of 0).

Creating a profile

TRMNL → Profiles → New

Identity

Field	Description
Name	Label (e.g. “Office calendar”).
Device	Target TRMNL.
Active	Inactive profiles are ignored for display.

Data source

Field	Description
Odoo Model	Business model to display (wizards and technical models are hidden).
View Type	Layout; options depend on the model (see below).
Users (optional)	Affects Assigned to Me (user_id filter) and uid / user in Domain Filter evaluation. Empty = current Odoo user when you click Render Preview ; on device polls the environment user is often Public , so set Users for production profiles. Does not change calendar week/month timezone (that follows the rendering user’s timezone setting).

Click **Render Preview** (header) to generate the first PNG without waiting for a device poll.

View types (layouts)

View Type	Use for	Key options
List	List of records	Fields to Display first field = title, others = metadata; Max Records; Sort Order
Kanban	Cards grouped by stage	Kanban Stage Field; Fields to Display — first field = card title
Calendar	Month or week calendar	Calendar View (Month / Week), Week Mode, Reference (Current / Custom Date)
Graph	Bar or line chart	Graph Type (Bar / Line), group-by / measure fields, Chart Title, sort and limits

View types not supported by the chosen model are not offered (e.g. Calendar layout only for models with a calendar view in Odoo).

Calendar Settings (Calendar layout only)

Field	Options
Calendar View	Month, Week
Week Mode	Work Week (Mon–Fri), Full Week (Mon–Sun); week view only
Reference	Current (today), Custom Date
Reference Date	Shown when Reference = Custom Date

Calendar notes:

- **Quick Filter** is hidden in the UI for calendar layouts; the month/week window comes from **Calendar Settings** and **Reference** only. **Domain Filter** still applies. **Assigned to Me** still applies to calendar loads when configured (via user_id on events).
- **Month — Reference = Current** uses the **server date** (date.today()), not the profile **Users** timezone. **Week — today** highlight uses the rendering user's timezone (env.user.tz).

- **Week view** includes only **non-all-day** timed events, drawn between **07:00 and 19:00** (local time after UTC conversion using the rendering user's timezone). All-day events are **not** shown. Events outside the hour grid are clipped on screen.
- **Month view** indexes events by **start** day (UTC date from start before local conversion in the mapper); multi-day events appear on the start day only. Only a limited number of events fit per cell (**+N more** for overflow).
- **Max Records** applies to calendar event search as well (same field as list/kanban; calendar loader falls back to **200** only when the limit is unset/zero).

Graph Settings (Graph layout only)

Bar chart

- Group By field(required)
- Measure (blank = count records),
- Sort order
- Max Bars (default 10, max 20)

Line chart

- Date Field (required)
- Measure (blank = count)
- Line Date Group By (Day / Week / Month)
- Max Points (default 12, max 52)

Chart Title: optional; defaults to the group-by or date field label.

Filters and sorting

Field	Description
Quick Filter	No Filter, Assigned to Me, Today, This Week, This Month, Overdue (not used for the calendar date window)
Domain Filter	Extra Odoo domain (AND with quick filter). Supports uid, context_today(), True, False
Sort Order	e.g. date_deadline desc, name asc (not for graph layout)
Max Records	Cap for list/kanban ORM search (default 20). Calendar event search uses the same field, with a code fallback of 200 when loading calendar.event. Not shown for graph layout.

Preview and delivery

- After **Render Preview** or an automatic render on poll:
- The **Preview** section shows the PNG in the form (same bytes the device downloads).
- **Device Delivery Status** shows **Device Last Polled**, **Next Expected Poll**, **Device Refresh Rate** (stored value in **seconds**, read-only), and **Image URL** (when a reachable URL could be built).
- The footer on the final PNG shows last poll time in **Europe/Zurich** (hard-coded in the renderer).

Refresh rate vs render interval

Setting	Where	Meaning
Refresh Rate	TRMNL → Devices	How often the device polls /api/display (1–30 minutes; default 1).
Render Interval	Optional column Render Interval (min) on the profile list	How often Odoo regenerates the PNG (default 10 minutes) if data has not changed. Enable the column via the list view column picker.

The device may poll every minute but still show the same PNG until a re-render is triggered (see below).

Render Preview

June 2026

Mon	Tue	Wed	Thu	Fri	Sat	Sun
1	2 06:30 Meeting 1	3 05:45 Meeting 2	4 08:00 Meeting 3	5	6	7
8	9	10	11	12	13	14
15	16	17	18 08:00 Meeting 4	19	20	21
22	23	24	25	26	27	28
29	30					

Display - Last poll: 2026-06-01 13:34

DEVICE DELIVERY STATUS

Device Last Polled ?

Jun 1, 1:34 PM

Next Expected Poll

Jun 1, 1:35 PM

Device Refresh Rate ?

60

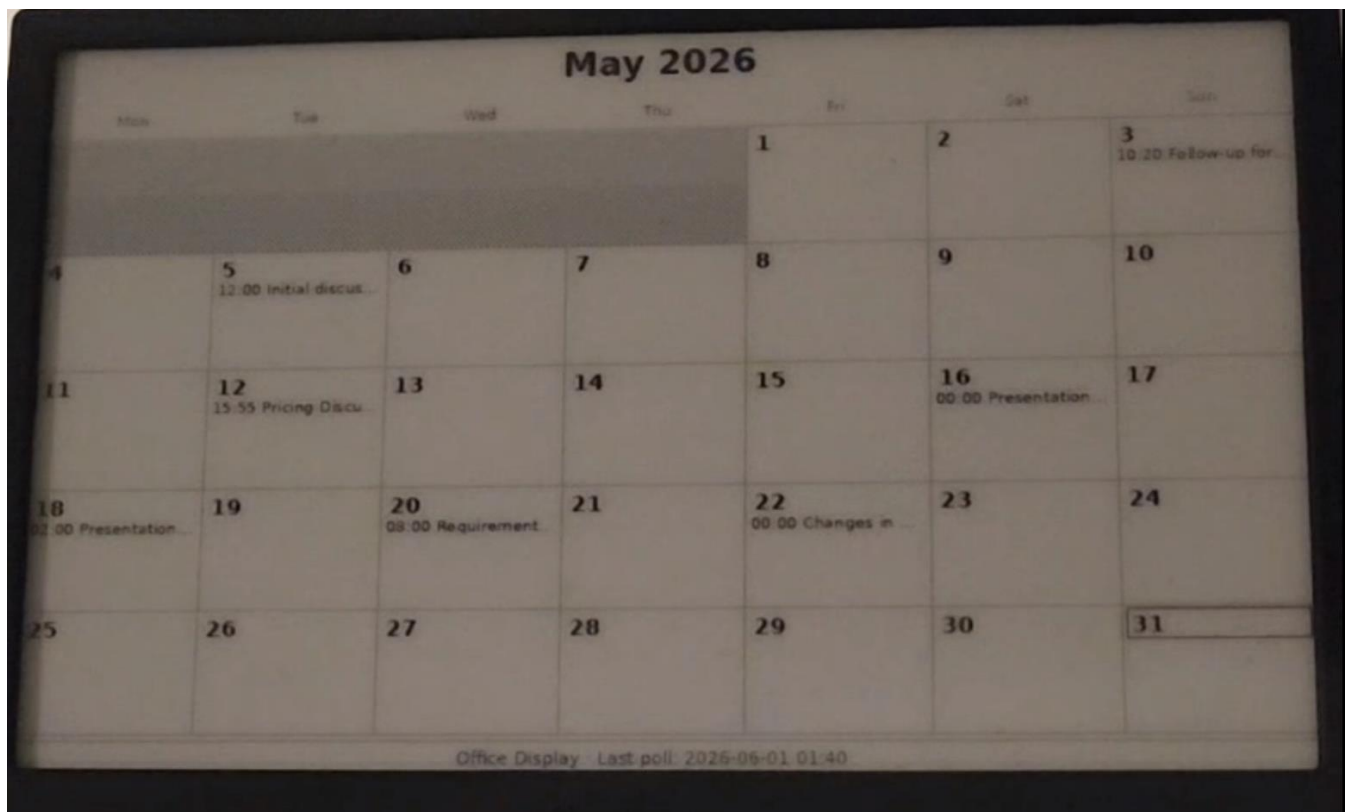
Image URL

<http://192.168.1.127:8069/api/profile/image/1>

Image URLs and web.base.url

TRMNL devices download images over HTTP. URLs must be reachable from the device's Wi-Fi network.

Situation	What happens
web.base.url is a LAN or public address the device can reach	Image URLs work; no warning.
web.base.url is localhost / loopback	Yellow URL Warning on the profile. After the first successful poll , Odoo may set or sync trmnl.public_base_url from the Host header.
trmnl.public_base_url is set	Settings → Technical → System Parameters . Used as the base for image URLs; warning suppressed.



Recommended before pairing (Docker / local):

1. **Settings** → **Technical** → **System Parameters**
2. Set **web.base.url** to your host's LAN URL, e.g. <http://192.168.1.50:8069>
3. Optionally set **trmnl.public_base_url** to the same value

On **Odoo.sh**, use your instance HTTPS URL; the device must reach it over the internet.

Profile image URLs include an **access_token** query parameter so only registered devices can download PNGs. Administrators can open the same URL when logged into Odoo without the token.

When Odoo re-renders the image

A new PNG is built when any of the following is true (on the next poll, or when you click **Render Preview**):

1. No preview exists yet
2. Source records changed (create / edit / delete on the profile's model)
3. **Render interval** elapsed (default 10 minutes)
4. The **trmnl** module was upgraded (renderer version changed)

The device **poll interval** is separate: if the device polls every 15 minutes, it still receives the **latest stored** PNG even if that PNG was rendered 5 minutes ago.

End-to-end workflow (example)

1. Pair the device (see **Device pairing**) → **Accepted** after `/api/setup`, default Abilium screen.
2. **TRMNL** → **Profiles** → **New** → model **Calendar Event**, **View Type Calendar**, **Device** = your TRMNL.
3. **Render Preview** → confirm the preview in the form.
4. Wait for the next device poll (or set **Refresh Rate** to **1** minute on the device for testing).
5. The display updates to your calendar PNG.

Troubleshooting

Device does not appear in Odoo

Check	Action
Custom Server URL	Must match Odoo's reachable address (LAN IP or Odoo.sh HTTPS), not localhost.
Soft Reset	Clear old credentials before Custom Server .
Wi-Fi	Device must join your network; reconnect to the TRMNL AP if needed.
Setup completed	Device should call /api/setup once; check Last Seen At on TRMNL → Devices .
Firewall	Allow inbound HTTP(S) to Odoo from the device subnet.

Device shows “Unauthorized” or Abilium error image

Check	Action
Display Policy	Default is Respond with an error until you Accept Device .
Status	Unknown Device or Token Mismatch → Accept Device if a token was stored.
Wrong server	Factory-reset device, re-enter Custom Server URL.
Token mismatch	Accept again or Reset Device and re-pair.

Profile preview works but device image does not update

Check	Action
URL Warning on profile	Fix web.base.url / trmnl.public_base_url; poll once for auto-sync.

Image URL empty	Device keeps previous filename; fix base URL.
Wrong profile	Use lowest sequence among active profiles for that device.
Refresh Rate	Lower to 1 minute for testing; press device back button to refresh.

Empty or wrong data on the display

Check	Action
Quick Filter / Domain Filter	Too restrictive → widen or clear domain.
Users on profile	Affects Assigned to Me and uid in domains.
Max Records	List/kanban default 20; increase if rows are missing.
View Type	Calendar layout is intended for Calendar Event .
Render Preview	Validates Odoo rendering before blaming the device.

Calendar-specific

Symptom	Likely cause
Event missing in week view	All-day event (excluded), outside 07:00–19:00 , or filtered out.
Event on wrong day	Month/week use server or rendering-user timezone; set the Odoo user's Timezone preference (or use Custom Date). Users on the profile does not fix calendar TZ.
Calendar layout empty list	Model is not Calendar Event — only that model uses the calendar renderer.
“+N more” on month cells	More events than fit in the cell.

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Display Policy: factory reset loop

If the device keeps resetting, **Display Policy** may be **Send a reset command**. Change to **Respond with an error** or **Automatically accept**, then re-pair.

Logs and debugging

- **TRMNL** → **Devices** → device form → **Logs** tab.
- Odoo server log: search for TRMNL during setup and display polls.
- Optional: **trmnl.api_debug** = True in system parameters for verbose API logging.

Module uninstall fails

If uninstall errors mention missing columns on trmnl_profile (for example active), use a build that includes the data-watcher uninstall fix, then retry uninstall.